

- 2 (a) Planets have been observed orbiting a star in another solar system. Measurements are made for the orbital radius r and the time period T of each of these planets.

The variation with r^3 of T^2 is shown in Fig. 2.1.

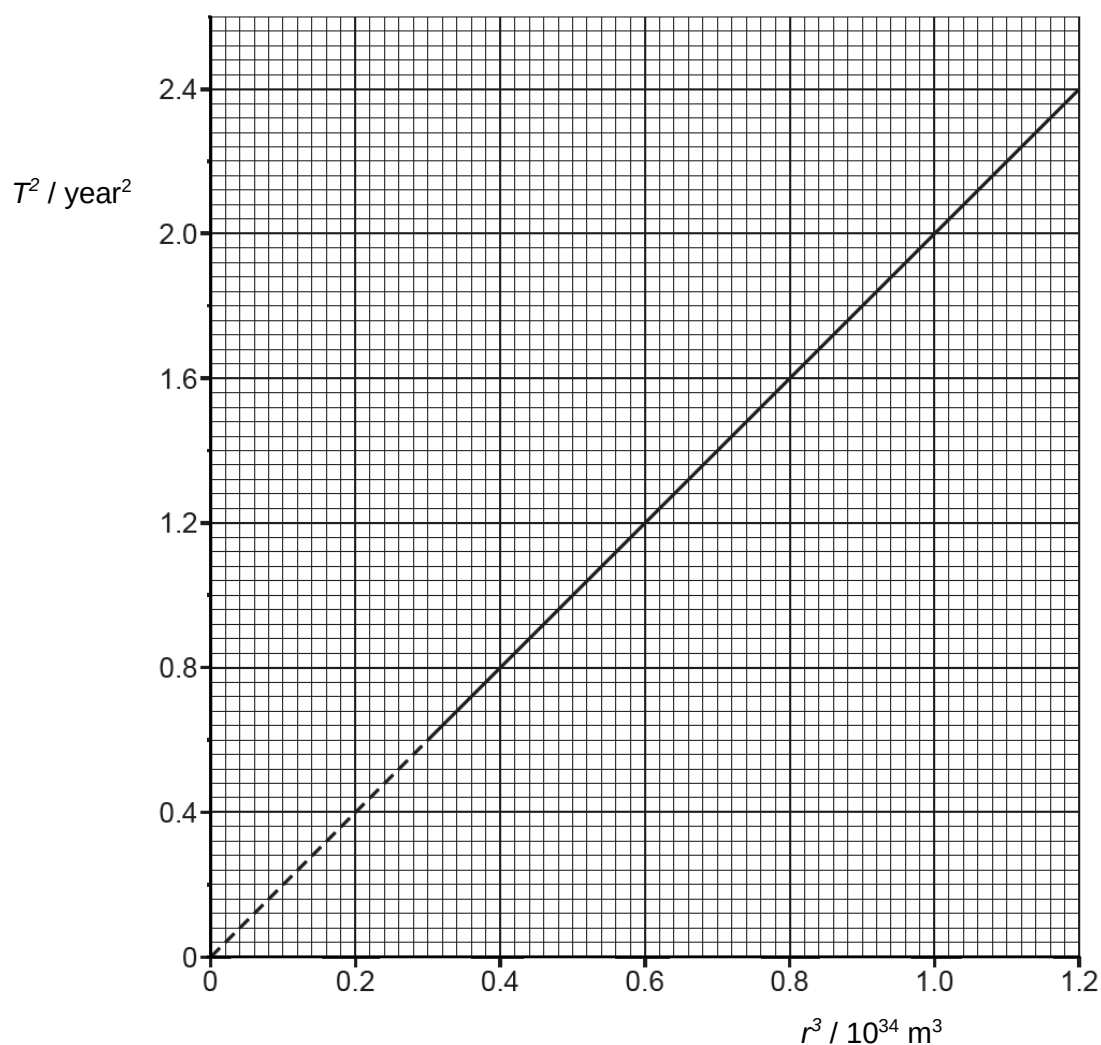


Fig. 2.1

The relationship between T and r is given by

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

where G is the gravitational constant and M is the mass of the star.

- (i) Determine the mass M of the star.

$$\text{Gradient} = \frac{2.4 - 0}{(1.2 - 0) \times 10^{34}} = 2.0 \times 10^{-34}$$

$$\frac{4\pi^2}{GM} = 2.0 \times 10^{-34} \times (365 \times 24 \times 60 \times 60)^2 \quad [1]$$

$$M = 2.98 \times 10^{30} \text{ kg or in 2 sf} \quad [1]$$

$$M = \dots\dots\dots \text{ kg} \quad [2]$$

- (ii) The radius of the star is 700 000 km. Determine the minimum speed with which gas particles from its surface have to be ejected to just escape from the star's pull of gravity.

By Conservation of Energy,

Initial KE + initial GPE at surface = Final KE + final GPE at infinity

$$\frac{1}{2} m v_{\text{escape}}^2 + (-GMm/r) = 0$$

$$v_{\text{escape}} = (2GM/r)^{1/2} = 7.53 \times 10^5 \text{ m/s}$$

$$\text{minimum speed} = \dots\dots\dots \text{ m s}^{-1} \quad [2]$$

- (iii) Hydrogen gas, consisting of hydrogen-2 particles, may be assumed to be an ideal gas. If the surface temperature of the star is 6000 K, determine whether hydrogen gas

(Find speed of hydrogen-2 particles)

$$\frac{1}{2} m v^2 = 3/2 kT$$

$$\frac{1}{2} (2 \times 1.66 \times 10^{-27}) v^2 = 3/2 (1.38 \times 10^{-23})(6000) \quad [1]$$

$$v = 8630 \text{ ms}^{-1} < v_{\text{esc}} = 7.53 \times 10^5 \text{ m/s}^{-1}, \text{ particles cannot escape} \quad [1]$$

Or

(Find initial KE required to escape)

$$\text{KE required} = \frac{1}{2} m v_{\text{escape}}^2 = \frac{1}{2} (2 \times 1.66 \times 10^{-27})(7.53 \times 10^5)^2 = 9.41 \times 10^{-16} \text{ J}$$

$$\text{KE of gas particles} = 3/2 (1.38 \times 10^{-23}) (6000) = 1.242 \times 10^{-19} \text{ J} \quad [1]$$

Since KE of particles is less than that required, particles cannot escape. [1]

- (iv) Some gas particles have very large kinetic energy to be able to escape from the star.

Given that the star is rotating about an axis through its poles, suggest why the gas particles at the equator of the star are more likely to escape the surface than those at the poles.

Gas particles at the equator is furthest away from this axis of rotation/largest radius of rotation. [1]

Hence particles at the equator has the largest speed of rotation [1]
and may exceed the minimum speed to escape the star's surface

(b) A satellite of mass m is also in orbit around the star in **(a)**. The radius of the orbit is r .

(i) Show that the kinetic energy E_k of the satellite is given by

$$E_k = \frac{GMm}{2r}.$$

Gravitational force by star on satellite provides for the centripetal force on the satellite

By Newton's 2nd Law,

$$\Sigma F = ma_c$$

$$GMm/r^2 = mv^2/r$$

$$\frac{1}{2} mv^2 = \frac{1}{2} GMm/r$$

[1]

(ii) On Fig. 2.2, sketch graphs to show the variation with orbital radius r of the

1. gravitational potential energy of the satellite. Label the graph U.
2. kinetic energy of the satellite. Label the graph K.
3. total energy of the satellite. Label the graph T.

[3]

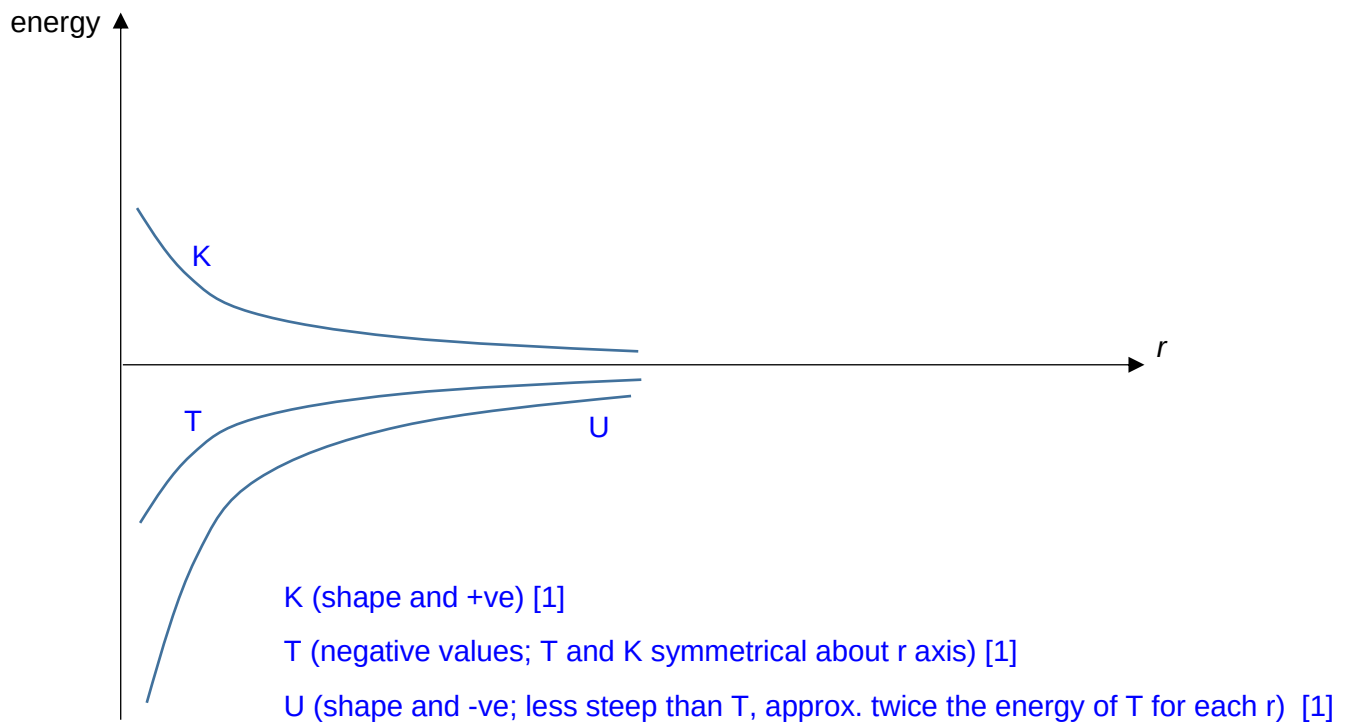


Fig. 2.2

[Total: 12]

- 3 A roller coaster ride in an amusement park consists of an unpowered car moving freely along a